



Tire Flat-Spot Test

1 Scope

Note: Nothing in the standard supersedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This document describes the procedure for testing passenger car and light truck tires for flat-spot.

1.2 Foreword. Increased levels of tire non-uniformity caused by a tire flat-spotting may contribute to the vibrations experienced by the customer. Flat-spot test results may be used to evaluate a tire against component flat-spot specifications or it may also be used to help diagnose vehicle vibration issues.

1.2.1 Flat-spot testing measures the change in tire uniformity caused by extended periods of static loading at a defined temperature. It quantifies both the initial and permanent effects of a flat-spot.

1.2.2 The flat-spot result requires two high-speed uniformity measurements. The first measurement is taken before the static load application and is defined as initial time-based uniformity. The second is taken after the static load application and is defined as post flat-spot time-based uniformity. The flat-spot data is the decay of the vector differences between the uniformity measurements at equal times, for a period of 30 minutes.

1.2.3 The first two harmonics of radial (Fz) and Fore/aft (Fx) are of the forces of primary interest.

1.3 Applicability. These procedures apply to the standard flat-spot testing of passenger car and light truck tires. Uniformity is measured on a Highway Speed Uniformity (HSU) machine at zero slip angle and zero inclination angle.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

SAE J332

2.2 GM Standards/Specifications.

None

2.3 Additional References.

GM Global Tire Database

3 Resources

3.1 Facilities. Not applicable.

3.2 Equipment. All details associated with the test apparatus are included in SAE J332 Testing Machines for Measuring the Uniformity of Passenger Car and Light Truck Tires.

3.2.1 The roadwheel should be covered with either medium grit 3M Safety Walk or 80 grit Three-M-ite. The surface should be replaced at routine intervals to eliminate it as a source of testing variability.

3.3 Test Vehicle/Test Piece. A tire-wheel assembly is required to run a flat-spot test. The test requester will specify the width and rim flange of the wheel.

3.4 Test Time.

Calendar time	1.5 days
Test hours	26 hours
Coordination hours:	1 hour

3.5 Test Required Information. Required information includes test temperature, tire size, construction number, load rating, service description, and wheel width. Additional information required may vary by test site.

3.6 Personnel/Skills. Test operator should have adequate training and be familiar with the local operator's procedure.

4 Procedure

4.1 Preparation.

4.1.1 Tires must be mounted on the appropriate wheels. See 3.3 Test Vehicle/Test Piece.

4.1.2 All tires must be mounted in the proper orientation with respect to the front face of the test wheels. Whitewall tires must be mounted whitewall out with respect to the front face of the wheels. Blackwall tires must be mounted with the cosmetic

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

or intended outboard sidewall mounted out with respect to the front face of the test wheels.

4.1.3 Assemblies or individual tires and wheels should be at an ambient room temperature of $24.0 \pm 3.0^\circ\text{C}$ for a minimum of 2 h before a flat-spot test.

4.1.4 The assembly should always be mounted on the machine in the standard position, in which the valve stem is aligned with the zero mark on the spindle. If this is not possible because the spindle studs do not line up with the boltholes, the assembly is rotated clockwise until the studs and boltholes do line up. The angle through which the assembly was rotated is then used to correct the phase angles of the Fast Fourier Transform (FFT).

4.1.5 Minimizing the radial runout is very important to minimize mounting contribution to uniformity. Follow local mounting instructions using expandable collets and other mounting hardware in order to optimize the centering of the machine mounting.

4.2 Conditions.

4.2.1 Environmental Conditions. Ambient temperature of the uniformity machine test area must be kept at $24.0 \pm 3.0^\circ\text{C}$. A method for monitoring ambient temperature must be in place.

4.2.2 Test Conditions. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 Load and Pressure. Tires must be tested at standard test conditions unless otherwise specified by the tire test request. Standard test conditions are defined in Table 1.

Table 1: Calculation of Standard HSU Test Conditions

Tire Type	Load Range	Test Load	Test Pressure
Passenger Car	All	70% of max load	210 kPa
Light Truck or Commercial	All	70% of load capacity at 350 kPa	350 kPa

4.2.2.2 Speed. The standard test speed used to evaluate flat-spot performance is 120 km/h in the clockwise direction.

4.3 Instructions. The flat-spot test result requires two high-speed uniformity measurements, one before the stationary load application (initial time-

based uniformity) and one after (post flat-spot time-based uniformity).

4.3.1 Acquiring Initial Time-Based Uniformity.

4.3.1.1 To acquire the initial time-based uniformity, no break-in or warm-up is used.

4.3.1.2 Because the initial flat-spot decays quickly, the following steps should be used to reach the test conditions, with the interest of exercising the tire as little as possible before data acquisition.

4.3.1.2.1 With the roadwheel stationary, load the tire to 500 N.

4.3.1.2.2 Accelerate the roadwheel to 120 km/h as quickly as possible.

4.3.1.2.3 Adjust the tire load to the Test Load.

4.3.1.2.4 Steady-State conditions must be reached within 90 s of road-to-wheel contact. Within 60 s is preferred.

4.3.1.3 Begin data acquisition as soon as steady-state conditions are met. Data points should be gathered at 30 s intervals (60 s intervals are allowed). Continue capturing data for 30 minutes, where time zero begins once the tire has reached the steady-state test conditions. As a current GM practice, eight consecutive revolutions of data are captured every 30 s for a period of 30 minutes. This group of 60 data sets is called the initial time-based uniformity

4.3.2 Inducing the Flat-Spot. Immediately after the first data collection is completed (< 10 minutes), place the assembly on a loading fixture in a climate controlled room at the correct test temperature. Keep the assembly loaded at the test load and test temperature for the specified times.

4.3.2.1 Location. The location for the flat-spot load is not specified. Flat-spot results have been shown to be independent of the flat-spot location.

4.3.2.2 Method. The load control method is the preferred method of inducing the flat-spot. Deflection control may be acceptable if the temperature verses deflection/load relationship is known and can be accommodated for sufficiently.

4.3.2.3 Temperature and Duration. The Standard Flat-spot Test Temperature is the 43.3°C (110°F) test. Other temperature conditions may also be run as required. The three temperature conditions defined for testing flat-spot are described below.

4.3.2.3.1 Standard. The Standard Test Temperature is as follows:

- $43.3 \pm 2.8^\circ\text{C}$ ($110 \pm 5^\circ\text{F}$) for 22 h, then
- $21.1 \pm 2.8^\circ\text{C}$ ($70 \pm 5^\circ\text{F}$) for 2 h

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

In the case of the standard test procedure of 43.3°C (110°F) temperature, the tire is first loaded and in the 43.3°C (110°F) environment for 22 h. The tire is then moved to ambient temperature conditions of $21.1 \pm 2.8^\circ\text{C}$ ($70 \pm 5^\circ\text{F}$) while maintaining the applied load at the same location on the tire. The tire is allowed to cool for two hours in the ambient temperature environment.

4.3.2.3.2 Ambient Temperature. An ambient temperature test of $21.1 \pm 2.8^\circ\text{C}$ ($70 \pm 5^\circ\text{F}$) would have the tire loaded and in the ambient temperature environment for the full 24 h.

4.3.2.3.3 Cold Temperature. A cold temperature test of $1.1 \pm 2.8^\circ\text{C}$ ($30 \pm 5^\circ\text{F}$) would have the tire loaded and in the cold temperature environment for the full 24 h.

4.3.3 Acquiring Post Flat-Spot Time-Based Uniformity.

4.3.3.1 Transfer the test assembly from the loading fixture to the uniformity machine. It is important to complete the transfer quickly (< 10 minutes), avoiding extraneous loading and handling.

4.3.3.2 When acquiring the post flat-spot time-based uniformity measurements, there is again no break-in or warm-up period.

4.3.3.3 Because the initial flat-spot decays quickly, the following steps should be used to reach the test conditions, with the interest of exercising the tire as little as possible before data acquisition.

4.3.3.3.1 With the roadwheel stationary, load the tire to 500 N.

4.3.3.3.2 Accelerate the roadwheel to 120 km/h as quickly as possible.

4.3.3.3.3 Adjust the tire load to the Test Load.

4.3.3.3.4 Steady State conditions must be reached within 90 s of road-to-wheel contact. Within 60 s is preferred.

4.3.3.4 Begin data acquisition as soon as steady state conditions are met. Data points should be gathered at 30 s intervals (intervals of 60 s are allowed). Continue capturing data for 30 minutes, where time zero begins once the tire has reached the steady-state test conditions. As a current GM practice, eight consecutive revolutions of data are captured every 30 s for a period of 30 minutes. This group of 60 data sets is called the post flat-spot time-based uniformity.

Note: The unbalance data is **not** captured at the end of the 30 minutes in flat-spot testing as is done in HSU.

5 Data

5.1 Calculations. Radial (F_z) and Fore/aft (F_x) are the forces of primary interest for flat-spot.

Note: These same calculations can be applied to any of the measured forces and moments.

Individual data sets, within both initial time-based uniformity and post flat-spot time-based uniformity are analyzed in the following manner:

5.1.1 All eight revolutions of the acquired data are averaged into one representative revolution.

5.1.2 The magnitudes and phase angles of radial and fore/aft forces are calculated based on the following formulations.

5.1.2.1 Radial.

$$F_z(i) = \sum_{j=1}^n R_j H * \cos\left(\frac{2 * \pi I}{j * T} * \frac{i}{m} + R_j P\right)$$

Where:

$F_z(i)$ is measured as an 8-rev-averaged radial force in SAE coordinate system at each sample points in one revolution

i is variable index of sample points $i = 1, 2, \dots, m$

m is number of sample points per revolution, such as 64 256.

j is variable index of harmonic $j = 1, 2, \dots, n$

n is number of harmonic, such as 16, 32

$R_j H$ is decomposed magnitude (Newtons) of radial force at harmonic j^{TH} or frequency of $1/(j * T)$

$R_j P$ is decomposed phase angle (Degrees) of radial force at harmonic j^{TH} or frequency of $1/(j * T)$

T is time taken for tire to rotate one cycle at specific speed and load.

5.1.2.2 Fore/Aft.

$$F_x(i) = \sum_{j=1}^n F_j H * \cos\left(\frac{2 * \pi I}{j * T} * \frac{i}{m} + F_j P\right)$$

Where:

$F_x(i)$ is measured as an 8-rev-averaged fore/aft force in SAE coordinate system at each sample points in one revolution

$F_j H$ is decomposed magnitude (Newtons) of fore/aft force at harmonic j^{TH} or frequency of $1/(j * T)$

$F_j P$ is decomposed phase angle (Degrees) of fore/aft force at harmonic j^{TH} or frequency of $1/(j * T)$

5.1.3 Typically the first two harmonics are of primary interest for flat-spot.

5.1.4 Phase angle is always reported as a physical spot on the tire measured clockwise from the zero mark of the spindle. The phase angle is always

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

positive, so the n^{th} harmonic phase angle can be only from zero to 360 degrees /n.

5.1.5 Calculate the Flat-spot Results. The results of the initial time-based uniformity are subtracted from the results of post flat-spot time-based uniformity at each time step. A FFT will be performed on the difference between post flat-spot time-based uniformity and initial time-based uniformity to obtain both magnitudes and angles of force variations.

5.1.5.1 Polar charts are plotted for the resulting force vectors for the first two harmonics. Generally, the radial (Fz) and Fore/aft (Fx) vectors are of interest.

5.1.5.2 The magnitudes of time-based uniformity vector differences for a given parameter can be fit with an exponential decay function of the form:

$$Mag(\vec{U}_{corr}) = A + B \cdot e^{-Ct}$$

Where:

A is the permanent flat-spot

A+B is the initial flat-spot, and

C is the time constant where t has units of minutes.

5.1.5.3 The magnitudes of the time-based uniformity vector differences for each parameter of interest are generally plotted on a time scale and presented along with the exponential decay function above.

5.2 Interpretation of Results. The requester of the test will review the measured data, according to their specific needs.

5.3 Test Documentation. For Test Improvement Process (TIP) approved data being uploaded to the GM Global Tire Database, follow the electronic file formats in Appendix A; otherwise, document the test results as shown in the sample test report in Appendix B.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Applied Load: The force existing between the tire and the roadwheel during an HSU test.

Composite Magnitude: The peak-to-peak magnitude of the force variation over one complete tire rotation, 360 degrees.

Data:

Assembly Data: The difference between composite and unbalance data. Assembly data is calculated as composite minus unbalance.

Composite Data: The data acquired while the tire is run in a steady-state condition loaded against the roadwheel.

Flat-Spot: The tendency of a tire to change in uniformity when it is loaded statically for a period of time. The degree of flat-spotting is dependant on the duration for which the tire remains static and the conditions to which it is exposed, i.e. load, temperature, pressure, etc.

Unbalance Data: The data acquired during an HSU test, while the tire is free spinning, i.e. not touching the roadwheel.

Harmonic:

First Harmonic: The peak-to-peak magnitudes of the lowest or fundamental frequency of the force variation signal over one complete tire rotation of 360 degrees.

Inflation Pressure: The capped tire inflation pressure during an HSU test.

Machine Contribution: A measure of the machine's contribution to the non-uniformity that appears in the test data. Machine contributions can come from an out of balance spindle and eccentricities of the assembly when it is mounted to the machine. They are typically first harmonic phenomena, but they can appear in other harmonics. Machine contributions are different for every colleting/wheel construction combination.

Magnitude: The length of the vector that describes the force or moment of a harmonic of the tire. Force magnitudes are reported in Newtons. Moment magnitudes are reported in Newton meters. Magnitudes are always positive.

n^{th} Harmonic: The peak-to-peak magnitude of the frequency that is n times the lowest or fundamental frequency of the force variation signal over one complete tire rotation of 360 degrees.

Phase Angle: The angle of the vector that describes a harmonic of the tire. By GM standards, phase is reported in degrees as a physical spot on the tire, measured clockwise from the valve stem of the assembly. Phases are always positive. Thus, the first harmonic phase will always be from 0 to 360 degrees. The second harmonic phase will always be from 0 to 180 degrees. The n^{th} harmonic

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

phase will always be from 0 to 360 degrees/n. The 0 degrees reference marks the valve stem of the assembly when it is on a line segment connecting the center of the road wheel to the center of the assembly, as shown in Figure 1.

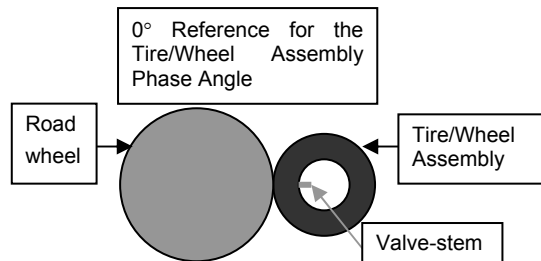


Figure 1: Phase Angle Reference

Speed: The speed of the tire during data acquisition.

Standard Position: The assembly is mounted on the machine such that the valve stem is aligned with the 0 mark on the spindle. If this is not possible because studs on the spindle do not match up with the bolt holes of the assembly, the assembly is rotated clockwise until the studs and bolt holes are aligned. The angle of rotation is noted so that it can be subtracted out of the phase angles from the FFT's. This is the mounting position for all HSU tests unless specified otherwise.

Vectors:

FxH: The Fore/Aft Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The fore/aft direction is also referred to as Fx in SAE coordinates.

LxH: The Lateral Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The lateral direction is also referred to as Fy in SAE coordinates.

MXxH: The Overturning Moment Variation Vector, magnitude and phase, for the x^{th} harmonic. The

overturning moment direction is also referred to as Mx in SAE coordinates.

MZxH: The Aligning Torque Variation Vector, magnitude and phase, for the x^{th} harmonic. The aligning torque direction is also referred to as Mz in SAE coordinates.

RxH: The Radial Force Variation Vector, magnitude and phase, for the x^{th} harmonic. The radial direction is also referred to as Fz in SAE coordinates.

7.2 Acronyms, Abbreviations, and Symbols.

CCW	Counter Clockwise
Coeff	Coefficient
CTS	Component Technical Specifications
CW	Clockwise
FFT	Fast Fourier Transform
FLTSPT	Flat Spot
HSU	Highway Speed Uniformity
Mag	Magnitude
OD	Outside Diameter
OE	Original Equipment
PDF	Portable Document Format
SAE	Society (of) Automotive Engineers
SSE	Sum (of) Squared Errors
TIP	Test Improvement Process
VTS	Vehicle Technical Specifications

8 Coding System

This standard shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:
Test to GMW15438

9 Release and Revisions

9.1 Release. This standard originated in October 2006. It was first approved by the Tire Global Subsystem Leadership team in November 2006. It was first published in March 2007.

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Appendix A

Table A1: Electronic File Format Definitions for GM Global Tire Database Submission

.TPC and .DAT File Naming Convention – Tire Flat-Spotting			
File Type	File Name	Explanation	Example
TPC	XXYMDDSS.TPC	XX - Supplier Code Y – Flat-spot M - Month (1 to 9 and A, B, C) DD - Day of Month SS - Test Number (ex. 01)	XXYB0426.TPC Flat-spot Test (Third Character = Y) Tested on November 4 (“B” = November, “04” = Test Day) Test Number 26

Table A2: TPC File Format

Line Number	Variable	Size	Example	Comments
General Test Information				
1	Data File Name ^{Note 1}	A14	Wyxxxxx.TPC	Name of this file ^{Note 1}
2	Data File Name ^{Note 1}	A14	TWYA0501.dat	
3	Report image File Name ^{Note 1}	A14	TWYA0501.pdf	PDF file of Excel report, use the filename of excel report file with .pdf extension
4	Test Number	A5	00005	Sequential numbers starting at 00001.
5	Tire Test Request No.	A8	00-421	Supplied by GM
6	Test Site	A3	xxx	Valid codes only (see GM Global Tire Database)
7	Tester	A3	xxx	Valid codes only (see GM Global Tire Database)
8	End Test Date	A10	07/19/2000	Month/Day/Year of final test point completion
9	Purpose/Comment	A50	Development	
10	Test Machine	A8	Dyno A	
11	Test Station	A8	Sta. 2	
12	Drum Diameter	F5.3	1.710	[m] Total OD including surface treatment thickness
13	Drum Surface	A12	3M 80 Grit	Description of test drum surface treatment
Tire Test Information				
14				Blank Line

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
15	Supplier	A3	xxx	Valid codes only (see GM Global Tire Database)
16	Brand Name	A20	Steeltex R4S	Name of tire model
17	Size	A13	P235/60R16	
18	Load Rating	A2	SL	Valid codes only (see GM Global Tire Database)
19	Service Description	A4	110Q	Load Index and Speed Rating ('Single' category for LT tires)
20	Service Des. (LT Dual)	A4	108Q	Load Index and Speed Rating of 'Dual' category for LT tires, if applicable
21	Sidewall Type	A5	BSW	Short description of sidewall type
22	Construction No.	A12	123456	
23	Tire ID	A14	105	Character string to identify tire or 'COMPOSITE' to identify composite data set.
24	DOT Serial	A6	xx1797	
25				Blank Line
26	Rim Width	F3.1	7.0	In [in] , to define rim width
27	Wheel Type	A9	OE	OE or Precision
28	Flat-spot Temperature1	A3	110	Primary Flat-spot temperature in °° F (70, 110)
29	Flat-spot Time1	A3	22	Time (hours) loaded at primary flat-spot Temperature
30	Flat-spot Temperature2	A3	70	Secondary flat-spot temperature in degrees F (e.g. cooling step for 110° test)
31	Flat-spot Time2	A3	2	Time (hours) loaded at secondary flat-spot temperature. (e.g. cooling step for 110° test)
32	Tested at the SRC?	A3	Yes	Yes or No. Is this test done at the standard reference load and pressure?
33	Wheel offset	F6.2	23.25	[mm] as necessary to define test requirement
34	Adapter Linear Offset	F6.2	23.25	[mm] as necessary to define test requirement

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
35	Adapter Phase Offset	I3	34	[deg] as necessary to define test requirement
36	Test Type	A3	FLTSPT	"FLT" for Flat-spot Test
Test Data				
37	Number of Test steps	I2	3	Number of sets of data information recorded. If run for 30 minutes and get data every 30 s, there are 60 sets.
38				Blank Line
39				Blank Line
40				Blank Line
41	Number of Harmonic Values	I2	4	Number of Harmonic values written to this file for each data point.
42	Date	A10	07/19/2000	Month/Day/Year of test point completion
43	Time	A8	15:13:55	Hour:Minute:Second of last test point completion (24 hour format)
44	Radial Load	I5	2148	[N] Test load
45	Inflation Pressure	I3	250	[kPa] Test inflation pressure
46	Speed	I3	120	[km/h] Road Wheel Speed
47	Rotation Direction	A3	CW	CW or CCW
48	Label	A80		"H_Index Coeff A Coeff B Coeff C Coeff D Coeff E Mag(T = 2) Mag(T = 30)"
Field 49 below is repeated for the number of harmonic values specified in field 41.				
49	Harmonic Index	I3	R1H	Coefficients of Flat-Spot Equation Harmonic index I = F1H, F2H, R1H, R2H
	Coeff A	F5.2	15.66	Coefficient A of the flat-spot exponential decay function
	Coeff B	F5.2	78.41	Coefficient B of the flat-spot exponential decay function
	Coeff C	F5.3	0.20	Coefficient C of the flat-spot exponential decay function
	Coeff D	F5.2	1.00	Populated only if using 5 coefficient equation. It should be blank for the 3 coefficient equation.
	Coeff E	F5.2	1.00	
	Mag (N), T = 2	I4	200	Magnitude(N) of Forces at time equals 2 minutes (as fit)

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
	Mag (N), T = 30	I4	30	Magnitude(N) of Forces at time equals 30 minutes (as fit)
	SSE	F4.1	6.2	Sum of Squared Errors when solving for the above coefficients.

Note 1: The file name must reflect the final test step date of completion of that tire.

Note: Unless specified otherwise, all values in fields are to be left-justified and the remaining field padded with spaces i.e. "00-421 ", not "00-421".

Note: Unless specified otherwise, each field is written to one line in the file.

Note: Empty line is to keep the same format as one in database interface.

Note: A DAT file must exist for every corresponding TPC file

Table A3: DAT File Format

Line Number	Variable	Size	Example	Comments
General Test Information				
1	Data File Name ^{Note 1}	A14	TWYxxxxx.DAT	Name of this file ^{Note 1}
2	Test Number	A5	00005	Sequential numbers starting at 00001.
3	Tire Test Request No.	A8	00-421	Supplied by GM
4	Test Site	A3	xxx	Valid codes only (see GM Global Tire Database)
5	Tester	A3	xxx	Valid codes only (see GM Global Tire Database)
6	End Test Date	A10	07/19/2000	Month/Day/Year of final test point completion
7	Purpose/Comment	A50	Development	
8	Test Machine	A8	Dyno A	
9	Test Station	A8	Sta. 2	
10	Drum Diameter	F5.3	1.710	[m] Total OD including surface treatment thickness
11	Drum Surface	A12	3M 80 Grit	Description of test drum surface treatment
Tire Test Information				
12				Blank Line
13	Supplier	A3	xxx	Valid codes only (see GM Global Tire Database)
14	Brand Name	A20	Steeltex R4S	Name of tire model
15	Size	A13	P235/60R16	
16	Load Rating	A2	SL	Valid codes only
17	Service Description	A4	110Q	Load Index and Speed Rating ('Single' category for LT tires)
18	Serv. Desc. (LT Dual)	A4	108Q	Load Index and Speed Rating of 'Dual' category for LT tires, if applicable

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
19	Sidewall Type	A5	BSW	Short description of sidewall type
20	Construction No.	A12	123456	
21	Tire ID	A14	105	Character string to identify tire or 'COMPOSITE' to identify composite data set.
22	DOT Serial	A6	xx1797	
23				Blank Line
24	Rim Width	F3.1	7.0	[in] to define rim width
25	Wheel Type	A9	OE	OE or Precision
26	Flat-spot Temperature1	A3	110	Flat-spot temperature in degrees F (70, 110)
27	Flat-spot Time1	A3	22	Time (hours) loaded for flat-spot
28	Flat-spot Temperature2	A3	70	Secondary Flat-spot Step Temperature (e.g., cooling step for 110° test)
29	Flat-spot Time2	A3	2	Secondary Flat-spot Step Time (e.g., cooling step for 110 °test)
30				Blank Line
31	Wheel offset	F6.2	23.25	[mm] as necessary to define test requirement
32	Adapter Linear Offset	F6.2	23.25	[mm] as necessary to define test requirement
33	Adapter Phase Offset	I3	34	[deg] as necessary to define test requirement
34	Test Type	A3	FLTSPT	"FLT" for Flat-spot Test
Test Data				
35	Number of Test steps	I2	3	Number of sets of data information recorded. If run for 30 minutes and get data every 30 s, there are 60 sets.
36				Blank Line
37				Blank Line
38				Blank Line
39	Number of Harmonic Values	I2	4	Number of Harmonic values written to this file for each data point.
40	Date	A10	07/19/2000	Month/Day/Year of test point completion
41	Time	A8	15:13:55	Hour:Minute:Second of test point completion in 24 hour format
42	Radial Load	I5	2148	[N] Test load
43	Inflation Pressure	I3	250	[kPa] Test inflation pressure
44	Speed	I3	120	[km/h] Road Wheel Speed
45	Rotation Direction	A3	CW	CW or CCW

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
46	Label	A80		"Flat-spot time-based data (before fit)" (2 nd run subtracted from 1 st run)
47	Label	A80		"Data Point Time F1Hmag F1Hph R1Hmag R1Hph F2Hmag F2Hph R2Hmag R2Hph "
Field 50 is repeated for the number of test steps specified in field 35. This section is the time-based data, before fit, 2 nd run subtracted from the 1 st run.				
50	Data point	I2	12	Data points numbered sequentially 1 thru 60 for example.
	Time	A8	15:13:55	Hour:Minute:Second of test point completion (24 hour format)
	F1H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 1 st harmonic (raw)
	R1H (raw)	I4	209	Magnitude (N) of Radial Forces for the 1 st harmonic (raw)
	F2H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 2 nd harmonic (raw)
	R2H (raw)	I4	209	Magnitude (N) of Radial Forces for the 2 nd harmonic (raw)
51	Label	A80		"Pre Flat-spot time-based data" 1st run (if available)
52	Label	A80		"Data Point Time F1Hmag F1Hph R1Hmag R1Hph F2Hmag F2Hph R2Hmag R2Hph"
Field 53 is repeated for the number of test steps specified in field 35. This section is the time-based data of the 1 st run separately, if available.				
53	Datapoint	I2	12	Data points numbered sequentially 1 thru 60 for example.
	Time	A8	15:13:55	Hour:Minute:Second of test point completion (24 hour format)
	F1H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 1 st harmonic (raw)
	R1H (raw)	I4	209	Magnitude (N) of Radial Forces for the 1 st harmonic (raw)
	F2H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 2 nd harmonic (raw)
	R2H (raw)	I4	209	Magnitude (N) of Radial Forces for the 2 nd harmonic (raw)
54	Label	A80		"Post Flat-spot time-based data" 2nd run ,if available)
55	Label	A80		"Flat-spot time-based data (before fit)" (2 nd run subtracted from 1 st run)
56	Label	A80		"Data Point Time F1Hmag F1Hph R1Hmag R1Hph F2Hmag F2Hph R2Hmag R2Hph"

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Line Number	Variable	Size	Example	Comments
Field 57 is repeated for the number of test steps specified in field 35. This section is the time-based data of the 2nd run separately, if available.				
57	Data point	I2	12	Data points numbered sequentially 1 thru 60 for example.
	Time	A8	15:13:55	Hour:Minute:Second of test point completion (24 hour. format)
	F1H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 1 st harmonic (raw)
	R1H (raw)	I4	209	Magnitude (N) of Radial Forces for the 1st harmonic (raw)
	F2H (raw)	I4	209	Magnitude (N) of Fore-Aft Forces for the 2nd harmonic (raw)
	R2H (raw)	I4	209	Magnitude (N) of Radial Forces for the 2nd harmonic (raw)

Note 1: The file name must reflect the final test step date of completion of that tire.

Note: Unless specified otherwise, all values in fields are to be left-justified and the remaining field padded with spaces i.e. "00-421 ", not "00-421".

Note: Unless specified otherwise, each field is written to one line in the file.

Note: Empty line is to keep the same format as one in database interface.

GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Appendix B - Sample Test Report

Tire Flatspot Test Report

Page 1 of 2

Brand/Model Name: Brand / Model Name
 Size/LR/Serv Descr: P235/70R17 EL 108S
 Sidewall: BW
 Construction: XXXXXXXXX

Work Request: 06-xxx
 # of Tires: 2
 Rim Width: 7.00 in
 Purpose/Comment: DEV

Test Site/Tester: TWS / TWS
 Machine/Station: Dyno A / STA 2
 Drum Diameter: 1.71
 Drum Surface: 3M 80 Grit

Flatspot Regression Model	
$Mag(\bar{U}_{corr}) = A + B \cdot e^{-Ct}$	Where: A is the permanent flatspot A+B is the initial flatspot C is the time constant t has units of minutes

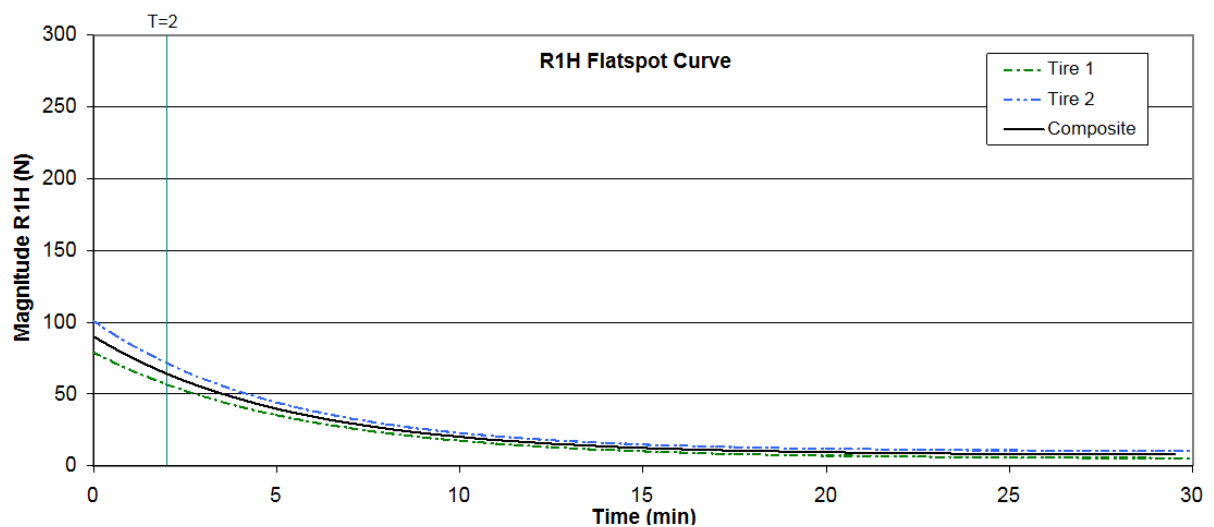
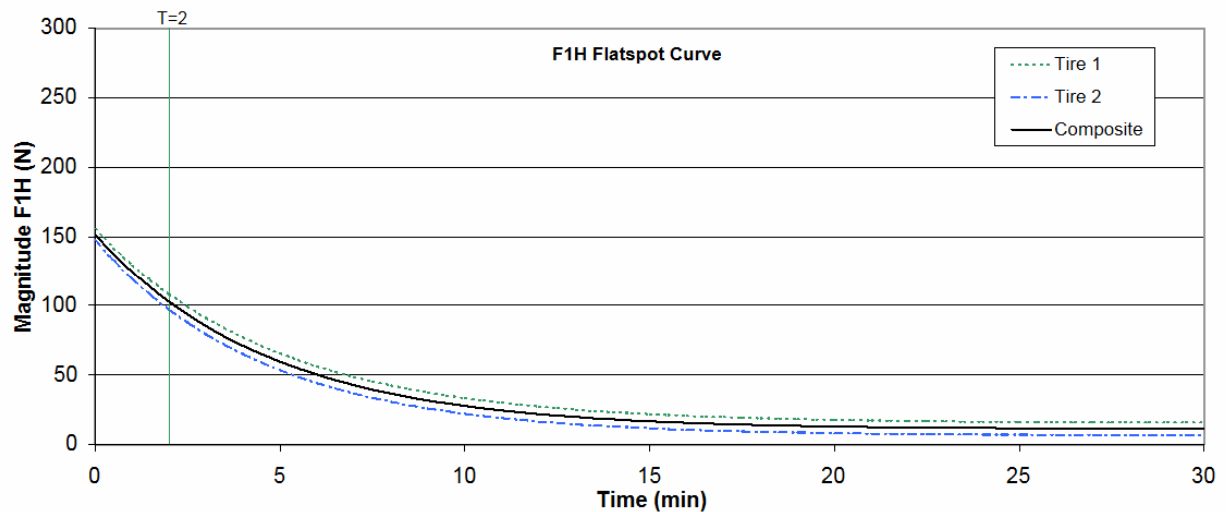
Flatspot Temperatures		
	Temp	Time
Primary	110	22
Secondary	70	2

Applied Correlation Equation
none

Test Conditions	
Load:	6350 N
Inflation:	260 kPa
Speed:	80 kph
Standard Y/N	Yes

1st Harmonic Results

1st Harmonic Results				F1H						R1H							
				Coefficients			Flatspot in (N) at:			Coefficients			Flatspot in (N) at:				
Dat File Name	Test Date	DOT	Tire ID	A	B	C		SSE	t = 2 min	t = 30 min	A	B	C		SSE	t = 2 min	t = 30 min
TWCA3002	10/30/2006	T74205	1	15.4	140.1	0.206		5.92	108	16	4.9	74.1	0.178		2.21	57	5
TWCA3001	10/30/2006	T74205	2	6.5	140.6	0.220		7.88	97	6	10.0	91.2	0.196		6.21	72	10



GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved

Appendix B - Sample Test Report continued

Tire Flatspot Test Report

Page 2 of 2

Brand/Model Name: Brand / Model Name
Size/LR/Serv Descr: P235/70R17 EL 108S
Sidewall: BW
Construction: xxxxxxxx

Work Request: 06-XXX
of Tires: 2
Rim Width: 7.00 in
Purpose/Comment: DEV

Test Site/Tester: TWS / TWS
Machine/Station: Dyno A / STA 2
Drum Diameter: 1.71
Drum Surface: 3M 80 Grit

Flatspot Regression Model
$Mag(\vec{U}_{corr}) = A + B \cdot e^{-\frac{t}{C}}$ Where: A is the permanent flatspot A+B is the initial flatspot C is the time constant t has units of minutes

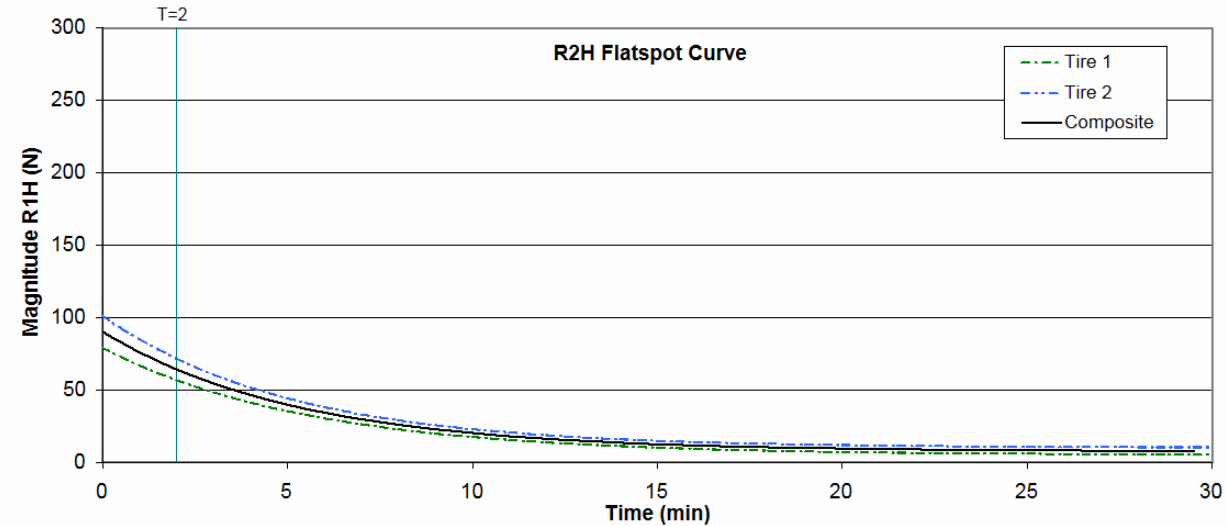
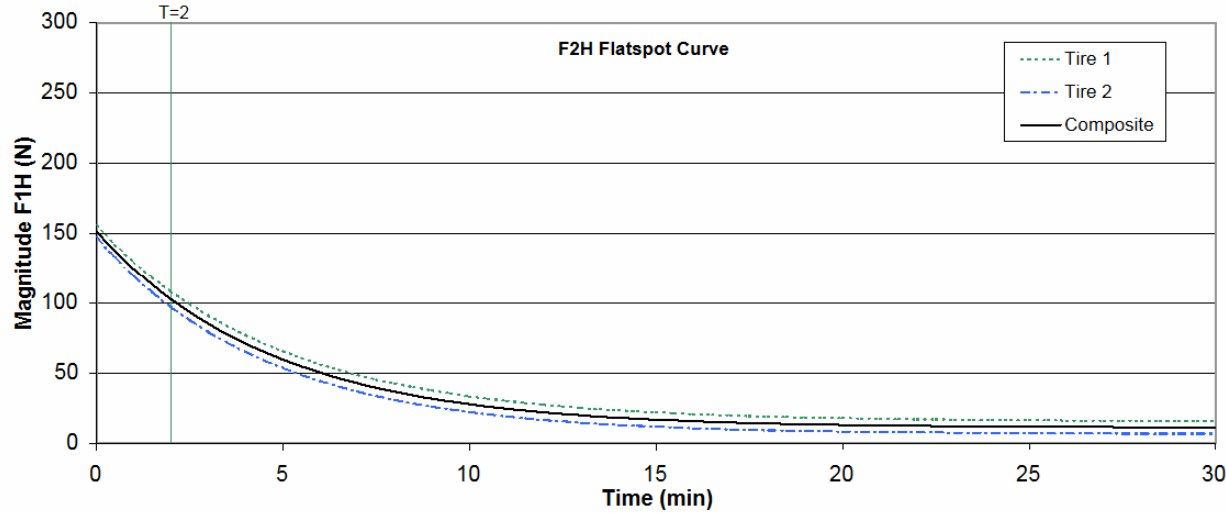
Flatspot Temperatures		
	Temp	Time
Primary	110	22
Secondary	70	2

Applied Correlation Equation
none

Test Conditions
Load: 6350 N
Inflation: 260 kPa
Speed: 80 kph
Standard Y/N: Yes

2nd Harmonic Results

Dat File Name	Test Date	DOT	Tire ID	F2H						R2H					
				Coefficients			Flatspot in (N) at:			Coefficients			Flatspot in (N) at:		
				A	B	C	SSE	t = 2 min	t = 30 min	A	B	C	SSE	t = 2 min	t = 30 min
TWCA3002	10/30/2006	XX4205	1	15.4	140.1	0.206	5.92	108	16	4.9	74.1	0.178	2.21	57	5
TWCA3001	10/30/2006	XX4205	2	6.5	140.6	0.220	7.88	97	6	10.0	91.2	0.196	6.21	72	10
Composite:				8.3	140.3	0.210	12.30	102	11	7.3	86.2	0.190	42.70	64	8



GM Confidential

© Copyright 2007 General Motors Corporation All Rights Reserved